

Last login: Tue Jun 23 09:03:24 on ttys001
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

tpch=# SET ROLE alice;

-- =====
-- STEP 8: Q2 using secure views
-- =====

EXPLAIN (ANALYZE, VERBOSE, BUFFERS)

```
select
    s_acctbal,
    s_name,
    n_name,
    p_partkey,
    p_mfg,
    s_address,
    s_phone,
    s_comment
from
    part,
    supplier_secure,
    partsupp_secure,
    nation,
    region
where
    p_partkey = ps_partkey
    and s_suppkey = ps_suppkey
    and p_size = 15
    and p_type like '%BRASS'
    and s_nationkey = n_nationkey
    and n_regionkey = region.r_regionkey
    and region.r_name = 'EUROPE'
    and ps_supplycost = (
        select
            min(ps_supplycost)
        from
            partsupp_secure,
            supplier_secure,
            nation,
            region
        where
            p_partkey = ps_partkey
            and s_suppkey = ps_suppkey
            and s_nationkey = n_nationkey
            and n_regionkey = region.r_regionkey
            and region.r_name = 'EUROPE'
    )
order by
    s_acctbal desc,
    n_name,
    s_name,
    p_partkey
limit 100;
SET
WARNING:  Log 1: Planner Input:
DETAIL:   {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname part
:colnames ("p_partkey" "p_name" "p_mfg" "p_brand" "p_type" "p_size"
"p_container" "p_retailprice" "p_comment")
}
:rtekind 0
:relid 16551
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname supplier_secure
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_ph
one" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
```

```

)
:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 16541
  :inh true
  :requiredPerms 2
  :checkAsUser 10
  :selectedCols (b 8 9 10 11 12 13 14)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  {fromlist (
    {RANGETBLREF
    :rtindex 1
    }
  )
}
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
  :opno 674
  :opfuncid 297
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 6
    :vartype 701
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
    }
    {FUNCEXP
    :funcid 316
    :funcresulttype 701
    :funcsetset false
    :funcvaradic false
    :funcformat 2
    :funccollid 0
    :inputcollid 0
    :args (
      {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
      }
    )
    :location -1
  )
}
:location -1
}
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
  {RANGETBLENTRY
  :alias
  {ALIAS
  :aliasname n
  :colnames <>
  }
  :eref
  {ALIAS
  :aliasname n
  :colnames ("n_nationkey" "n_name" "n_regionkey" "n_
  comment")
  }
  :rtekind 0
  :relid 16536
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCI true
  :securityQuals <>
  }
  {RANGETBLENTRY
  :alias
  {ALIAS
  :aliasname r
  :colnames <>
  }
  :eref
  {ALIAS
  :aliasname r
  :colnames ("r_regionkey" "r_name" "r_comment")
  }
  :rtekind 0
  :relid 16531
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 2
  :tablesample <>
  :lateral false
  :inFromCI true
  :securityQuals <>
  }
)
}
:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 16536
  :inh true
  :requiredPerms 2
  :checkAsUser 10
  :selectedCols (b 8 10)
  :insertedCols (b)

```

```

:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:location -1
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OPEXP
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0

```

```

        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
)
:override 0
:onConflict <>
:ReturningOldAlias <>
:ReturningNewAlias <>
:ReturningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname s_supplykey
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname s_name
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname s_address
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname s_nationkey
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}

```

```

    }
    :resno 5
    :resname s_phone
    :ressortgroupref 0
    :resorigtbl 16541
    :resorigcol 5
    :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 6
  :vartype 701
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 6
  :location -1
  }
  :resno 6
  :resname s_acctbal
  :ressortgroupref 0
  :resorigtbl 16541
  :resorigcol 6
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 7
  :vartype 25
  :vartypmod -1
  :varcollid 100
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 7
  :location -1
  }
  :resno 7
  :resname s_comment
  :ressortgroupref 0
  :resorigtbl 16541
  :resorigcol 7
  :resjunk false
  }
  }
  :override 0
  :onConflict <>
  :returningOldAlias <>
  :returningNewAlias <>
  :returningList <>
  :groupClause <>
  :groupDistinct false
  :groupingSets <>
  :havingQual <>
  :windowClause <>
  :distinctClause <>
  :sortClause <>
  :limitOffset <>
  :limitCount <>
  :limitOption 0
  :rowMarks <>
  :setOperations <>
  :constraintDeps <>
  :withCheckOptions <>
  :stmt_location -1
  :stmt_len -1
  }
  :security_barrier false
  :relid 17511
  :inh false
  :relkind v
  :rellockmode 1
  :perminfoindex 2
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
  {RANGETBLENTY
  :alias <>
  :eref
  {ALIAS
  :aliasname partsupp_secure
  :colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
s_comment")
  }
  }
  :rtekind 1
  :subquery
  {QUERY
  :commandType 1
  :querySource 0
  :canSetTag true
  :utilityStmt <>
  :resultRelation 0
  :hasAggs false
  :hasWindowFuncs false
  :hasTargetSRFs false
  :hasSubLinks true
  :hasDistinctOn false
  :hasRecursive false
  :hasModifyingCTE false
  :hasForUpdate false
  :hasRowSecurity false
  :hasGroupRTE false
  :isReturn false
  :cteList <>
  :rtable (
  {RANGETBLENTY
  :alias
  {ALIAS
  :aliasname ps
  :colnames <>
  }
  :eref
  {ALIAS
  :aliasname ps
  :colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplyco
st" "ps_comment")
  }
  }
  :rtekind 0
  :relid 16556
  :inh true
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
  )
  :rteperminfos (
  {RTEPERMISSIONINFO
  :relid 16556
  :inh true
  :requiredPerms 2
  :checkAsUser 10
  :selectedCols (b 8 9 10 11 12)
  :insertedCols (b)
  :updatedCols (b)
  }
  )

```

```

)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
):quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
){CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 ]
}
)
:location -1
}
){SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nati
onkey" "s_phone" "s_acctbal" "s_comment")
}
:rtkind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "
s_nationkey" "s_phone" "s_acctbal" "s_comment
")
}
:rtkind 0
:relid 16541
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
):rtperminfos (
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
)
):jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
):quals

```

```

{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:varctype 701
:varctymod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
{SUBLINK
:sublinktype 0
:sublinkid 0
:testexpr <>
:opername <>
:subselect
{QUERY
:commandtype 1
:querysource 0
:cansettag true
:utilitystmt <>
:resultrelation 0
:hasaggs false
:haswindowfuncs false
:haswindowfuncs false
:haswindowfuncs false
:hasdistincton false
:hasrecursive false
:hasmodifyingcte false
:hasforupdate false
:hasrowsecurity false
:hasgrouperte false
:isreturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name"
"n_regionkey" "n_comment")
}
:rtekind 0
:relid 16536
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:infromcl true
:securityquals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name"
"r_comment")
}
:rtekind 0
:relid 16531
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:infromcl true
:securityquals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredperms 2
:checkasuser 10
:selectedcols (b 8 10)
:insertedcols (b)
:updatedcols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredperms 2
:checkasuser 10
:selectedcols (b 8 9)
:insertedcols (b)
:updatedcols (b)
}
)
)
:jointree
{FROMEXP
:fromlist (
{RANGETBLENTY
:rindex 1
}
{RANGETBLENTY

```

```

:rtindex 2
}
)
:quals
{BOOLEXPR
:boolop and
:args (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 4
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
)
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
)
:location -1
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 0 6
9 85 82 79 80 69 ]
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>

```

```

        :returningList <>
        :groupClause <>
        :groupDistinct false
        :groupingSets <>
        :havingQual <>
        :windowClause <>
        :distinctClause <>
        :sortClause <>
        :limitOffset <>
        :limitCount <>
        :limitOption 0
        :rowMarks <>
        :setOperations <>
        :constraintDeps <>
        :withCheckOptions <>
        :stmt_location -1
        :stmt_len -1
    }
    :location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname s_supplekey
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname s_name
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname s_address
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname s_nationkey
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname s_phone
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0

```

```

:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname s_acctbal
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname s_comment
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 17511
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
(RTEPERMISSIONINFO
:relid 17511
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
)
)
):jointree
{FROMEXPR
:fromList (
{RANGETBLREF
:rtindex 1
}
)
):quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
:location -1
}
)
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
}
}

```

```

{FUNCEXP
:funcid 316
:funcresulttype 701
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname ps_partkey
:resortgroupref 0
:resorigtbl 16556
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname ps_suppkey
:resortgroupref 0
:resorigtbl 16556
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname ps_availqty
:resortgroupref 0
:resorigtbl 16556

```

```

:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varstype 701
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname ps_supplycost
:ressortgroupref 0
:resorigtbl 16556
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varstype 25
:varstypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname ps_comment
:ressortgroupref 0
:resorigtbl 16556
:resorigcol 5
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 17515
:inh false
:relkind v
:rellockmode 1
:perminfoindex 3
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 16536
:inh true
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 16531
:inh true
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16551
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17511
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17515
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16536
:inh true

```

```

:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:reid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
{RANGETBLREF
:rtindex 5
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 297
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 309
}
)
:location 307
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 332
}
{VAR
:varno 3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 2
:location 344
}
)
:location 342
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 367
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 376
:constvalue 4 [ 15 0 0 0 0 0 0 0 ]
}
)
:location 374
}
{OPEXP

```

```

:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 1
    :varattno 5
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 391
  }
  {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 403
    :constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ]
  }
)
:location 398
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 2
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 4
    :location 424
  }
  {VAR
    :varno 4
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 1
    :location 438
  }
)
:location 436
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 4
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 3
    :location 462
  }
  {VAR
    :varno 5
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 5
    :varattnosyn 1
    :location 476
  }
)
:location 474
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 5
    :varattno 2
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 5
    :varattnosyn 2
    :location 507
  }
  {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 523
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
  }
)
:location 521
}
{OPEXPR
:opno 670
:opfuncid 293
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0

```

```

:args (
{VAR
:varno 3
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location 544
}
{SUBLINK
:sublinkType 4
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname partsupp_secure
:colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps
_supplycost" "ps_comment")
}
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_supkey" "ps_availq
t" "ps_supplycost" "ps_comment")
}
}
:rtekind 0
:reldid 16556
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:reldid 16556
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtdindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 ]
}
)
}
:location -1
}
}

```

```

{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rttable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_a
ddress" "s_nationkey" "s_phone" "s_a
cctbal" "s_comment")
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rttable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name
" "s_address" "s_nationkey" "s
_phone" "s_acctbal" "s_comment
")
}
:rtekind 0
:relid 16541
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13
14)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0
0 0 0 0 ]
}
}
}
}
}
}

```

```

    }
  }
  :location -1
}
)
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nati
onkey" "n_name" "n
_regionkey" "n_com
ment")
}
:rtekind 0
:relid 16536
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regi
onkey" "r_name" "r
_comment")
}
:rtekind 0
:relid 16531
:inh true
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
}
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
}
:varlevelsup 0
:varreturningtype
0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
}
)
}

```

```

:varlevelsup 1
:varreturningtype
0
:varnosyn 1
:varattnosyn 4
:location -1
}
}
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b
)
:varlevelsup 0
:varreturningtype
0
:varnosyn 1
:varattnosyn 3
:location -1
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b
)
:varlevelsup 0
:varreturningtype
0
:varnosyn 2
:varattnosyn 1
:location -1
}
}
:location -1
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 2
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b
)
:varlevelsup 0
:varreturningtype
0
:varnosyn 2
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false

:location -1
:constvalue 10 [ 4
0 0 0 0 69 85 82 7
9 80 69 ]
}
}
:location -1
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false

:location -1
:constvalue 4 [ 1
0 0 0 0 0 0 ]
}
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}

```

```

:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname s_supkey
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname s_name
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname s_address
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname s_nationkey
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname s_phone
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname s_acctbal
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1

```

```

        :varattnosyn 7
        :location -1
    }
    :resno 7
    :resname s_comment
    :ressortgroupref 0
    :resorigtbl 16541
    :resorigcol 7
    :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 17511
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17511
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
):jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:location -1
}
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcsetset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
)

```

```

        :location -1
      }
    }
    :location -1
  }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname ps_partkey
    :ressortgroupref 0
    :resorigtbl 16556
    :resorigcol 1
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location -1
    }
    :resno 2
    :resname ps_suppkey
    :ressortgroupref 0
    :resorigtbl 16556
    :resorigcol 2
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 3
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location -1
    }
    :resno 3
    :resname ps_availqty
    :ressortgroupref 0
    :resorigtbl 16556
    :resorigcol 3
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 4
      :vartype 701
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 4
      :location -1
    }
    :resno 4
    :resname ps_supplycost
    :ressortgroupref 0
    :resorigtbl 16556
    :resorigcol 4

```

```

:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname ps_comment
:ressortgroupref 0
:resorigtbl 16556
:resorigcol 5
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 17515
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname supplier_secure
:colnames ("s_suppkey" "s_name" "s_address" "s_nationk
ey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable {
{RANGETBLENTY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_n
ationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos {
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
}
}
:jointree
{FROMEXPR
:fromlist {
{RANGETBLREF
:rteindex 1
}
}
}
:quals
{BOOLEXP
:boolop and
:args {
{OPEXP
:opno 674
:opfuncid 297
:oprresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args {
{VAR
:varno 1
:varattno 6
:varatype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0

```

```

:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 0 0 0 0 0 0 0 ]
  }
)
:location -1
}
)
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs false
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks false
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE false
    :isReturn false
    :cteList <>
    :rtable (
      {RANGETBLENTTRY
        :alias
          {ALIAS
            :aliasname n
            :colnames <>
          }
        :eref
          {ALIAS
            :aliasname n
            :colnames ("n_nationkey" "n_name" "n_
              _regionkey" "n_comment")
          }
        :rtekind 0
        :relid 16536
        :inh true
        :relkind r
        :rellockmode 1
        :perminfoindex 1
        :tablesample <>
        :lateral false
        :inFromCl true
        :securityQuals <>
      }
      {RANGETBLENTTRY
        :alias
          {ALIAS
            :aliasname r
            :colnames <>
          }
        :eref
          {ALIAS
            :aliasname r
            :colnames ("r_regionkey" "r_name" "r_
              _comment")
          }
        :rtekind 0
        :relid 16531
        :inh true
        :relkind r
        :rellockmode 1
        :perminfoindex 2
        :tablesample <>
        :lateral false
        :inFromCl true
        :securityQuals <>
      }
    )
    :rteperminfos (
      {RTEPERMISSIONINFO
        :relid 16536
        :inh true
        :requiredPerms 2
        :checkAsUser 10
        :selectedCols (b 8 10)
        :insertedCols (b)
        :updatedCols (b)
      }
      {RTEPERMISSIONINFO
        :relid 16531
        :inh true
        :requiredPerms 2
        :checkAsUser 10
        :selectedCols (b 8 9)
        :insertedCols (b)
        :updatedCols (b)
      }
    )
  )
}
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
      {RANGETBLREF
        :rtindex 2
      }
    )
  }
:quals
  {BOOLEXP
    :boolop and
    :args (
      {OPEXP
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 1
            :varattno 1
            :vartype 23
          }
        )
      }
    )
  }

```

```

        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
    }
    {VAR
        :varno 1
        :varattno 4
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 1
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 4
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 3
        :location -1
    }
    {VAR
        :varno 2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location -1
    }
)
:location -1
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
    {VAR
        :varno 2
        :varattno 2
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 2
        :location -1
    }
    {CONST
        :consttype 25
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 69 8
            5 82 79 80 69 ]
    }
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1

```

```

    }
    )
    :location -1
  }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname s_suppkkey
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname s_name
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname s_address
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname s_nationkey
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname s_phone
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varatype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname s_acctbal
:ressortgroupref 0
:resorigtbl 16541
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varatype 25
:vartypmod -1

```

```

        :varcolloid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 7
        :location -1
    }
    :resno 7
    :resname s_comment
    :ressortgroupref 0
    :resorigtbl 16541
    :resorigcol 7
    :resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 17511
:inh false
:relkind v
:rellockmode 1
:perminfoindex 2
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_com
ment")
}
:rtekind 0
:relid 16536
:inh true
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 16531
:inh true
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17515
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17511
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 840
  }
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 852
  }
)
:location 850
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location 891
  }
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location 903
  }
)
:location 901
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 2
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 4
    :location 942
  }
  {VAR
    :varno 3
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 1
    :location 956
  }
)
:location 954
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 3
    :location 996
  }
  {VAR
    :varno 4
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 1
    :location 1010
  }
)
:location 1008
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 4
    :varattno 2
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 2
    :location 1057
  }
  {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 1073
    :constvalue 10 [ 40 0 0 0 0 69 85 82 79 80 69 ]
  }
)
:location 1071
}
}
:location 887
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {AGGREF
      :aggfnoid 2136
      :aggtype 701
      :aggcollid 0
      :inputcollid 0
      :aggtranstype 0
      :aggargtypes (o 701)
      :aggdirectargs <>
      :args (
        {TARGETENTRY
          :expr
          {VAR
            :varno 1
            :varattno 4
            :vartype 701
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 4
            :location 613
          }
          :resno 1
          :resname <>
          :ressortgrouppref 0
          :resorigtbl 0
          :resorigcol 0
          :resjunk false
        }
      )
      :aggorder <>
      :aggdistinct <>
      :aggfilter <>
      :aggstar false
      :aggvariadic false
      :aggkind n
      :aggpresorted false
      :agglevelsup 0
      :aggssplit 0
      :aggno -1
      :aggtransno -1
      :location 609
    }
    :resno 1
    :resname min
    :ressortgrouppref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 560
}
}
:location 558
}
}
:location 328
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 2
      :varattno 6
      :vartype 701
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 2
      :varattnosyn 6
      :location 51
    }
    :resno 1
    :resname s_acctbal
    :ressortgrouppref 1
    :resorigtbl 17511
    :resorigcol 6
  }
)

```

```

:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 2
:location 70
}
:resno 2
:resname s_name
:ressortgroupref 3
:resorigtbl 17511
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 86
}
:resno 3
:resname n_name
:ressortgroupref 2
:resorigtbl 16536
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 102
}
:resno 4
:resname p_partkey
:ressortgroupref 4
:resorigtbl 16551
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 121
}
:resno 5
:resname p_mfgr
:ressortgroupref 0
:resorigtbl 16551
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 137
}
:resno 6
:resname s_address
:ressortgroupref 0
:resorigtbl 17511
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 5
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 5
:location 156
}
:resno 7
:resname s_phone
:ressortgroupref 0
:resorigtbl 17511
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 7
:location 173

```

```

    }
    :resno 8
    :resname s_comment
    :ressortgroupref 0
    :resorigtbl 17511
    :resorigcol 7
    :resjunk false
  }
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 670
    :sortop 674
    :reverse_sort true
    :nulls_first true
    :hashable true
  }
  {SORTGROUPCLAUSE
    :tleSortGroupRef 2
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
  {SORTGROUPCLAUSE
    :tleSortGroupRef 3
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
  {SORTGROUPCLAUSE
    :tleSortGroupRef 4
    :eqop 96
    :sortop 97
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount
  {FUNCEXP
    :funcid 481
    :funcresulttype 20
    :funcresultset false
    :funcvariadic false
    :funcformat 2
    :funccollid 0
    :inputcollid 0
    :args (
      {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location 1181
        :constvalue 4 [ 100 0 0 0 0 0 0 0 ]
      }
    )
    :location -1
  }
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

```

WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=481 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 proparallel=s (s=safe r=restricted u=unsafe)
WARNING: Log 2 : After pull up At starting of Path Generation:

```

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname partsupp_secure
        :colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
          s_comment")
      }
    :rtakind 1
    :subquery <>
    :security_barrier false
    :relid 17515
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 1
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname supplier_secure
        :colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
          s_acctbal" "s_comment")
      }
    :rtakind 1
    :subquery <>
    :security_barrier false
    :relid 17511
    :inh false
    :relkind v
  }
)

```

```

:rellockmode 1
:perminfoindex 2
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
:rtekind 0
:relid 16556
:inh false
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 17511
:inh false
:relkind v
:rellockmode 1
:perminfoindex 6
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 7
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 8
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 16531
:inh false

```

```

:relkind r
:rellockmode 1
:perminfoindex 9
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 10
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 11
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 12
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperinfos (
{RTEPERMISSIONINFO
:relid 17515
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17511
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16556
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17511
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
)

```

```

}
{RTEPERMISSIONINFO
:reld 16531
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:reld 16541
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:reld 16536
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:reld 16531
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 5
}
)
:quals (
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 3
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:rarg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 7
}
)
:quals (
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 8
}
{RANGETBLREF
:rtindex 9
}
)
:quals <>
}

```

```

:usingClause <>
:join_using_alias <>
:quals (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 8
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 8
        :varattnosyn 1
        :location -1
      }
      {VAR
        :varno 7
        :varattno 4
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 7
        :varattnosyn 4
        :location -1
      }
    )
    :location -1
  }
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 8
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 8
        :varattnosyn 3
        :location -1
      }
      {VAR
        :varno 9
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 9
        :varattnosyn 1
        :location -1
      }
    )
    :location -1
  }
  {OEXPR
    :opno 98
    :opfuncid 67
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 100
    :args (
      {VAR
        :varno 9
        :varattno 2
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 9
        :varattnosyn 2
        :location -1
      }
      {CONST
        :consttype 25
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
      }
    )
    :location -1
  }
)
:alias <>
:rtindex 0
)
:usingClause <>
:join_using_alias <>
:quals (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 7
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 7
        :varattnosyn 1
        :location -1
      }
      {VAR
        :varno 5
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
      }
    )
  }
)

```

```

:varnosyn 5
:varattnosyn 2
:location -1
}
)
:location -1
}
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 10
}
)
:quals (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 10
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 10
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 11
}
{RANGETBLREF
:rtindex 12
}
)
:quals <>
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 11
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 11
:varattnosyn 1
:location -1
}
{VAR
:varno 10
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 10
:varattnosyn 4
:location -1
}
)
:location -1
}
)
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 11
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 11
    :varattnosyn 3
    :location -1
  }
  {VAR
    :varno 12
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 12
    :varattnosyn 1
    :location -1
  }
)
:location -1
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 12
    :varattno 2
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 12
    :varattnosyn 2
    :location -1
  }
  {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
  }
)
:location -1
}
)
:alias <>
:rtindex 0
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
)
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {PARAM
    :paramkind 1
    :paramid 0
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location 840
  }
  {VAR
    :varno 5
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 5
    :varattnosyn 1
    :location -1
  }
)
:location 850
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 10
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 10
    :varattnosyn 1
    :location -1
  }
  {VAR
    :varno 5
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 5
    :varattnosyn 2
    :location -1
  }
)
:location 901
}

```

```

{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 19
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 10
:varattnosyn 4
:location -1
}
  {VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 956
}
)
:location 954
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 3
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 996
}
  {VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 1010
}
)
:location 1000
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
:varno 4
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 1057
}
  {CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 1073
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location 1071
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
:expr
  {AGGREG
:aggfnoid 2136
:aggtype 701
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes (o 701)
:aggdirectargs <>
:args (
  {TARGETENTRY
:expr
  {VAR
:varno 5
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 4
:location -1
}
}
:resno 1
:resname <>
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
}

```

```
WARNING: is_parallel_safe: INPUT qual = <<
WARNING: is_parallel_safe: OUTPUT qual = << | parallel_safe = true | reason = PROPARELL_SAFE
WARNING: is_parallel_safe: INPUT qual = { (VAR :varno 3 :varratno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1) :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1) }
WARNING: is_parallel_safe: OUTPUT qual = { (VAR :varno 3 :varratno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1) : (VAR :varno 3 :varratno 3 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 3 :location -1) | parallel_safe = true | reason = PROPARELL_SAFE
WARNING: is_parallel_safe: INPUT qual = { (RESTRICTINFO :clause (OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args { (VAR :varno 4 :varratno 2 :vartype 25 :vartypemod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1) (CONST :consttype 25 :consttypemod -1 :constcollid 100 :constlen -1 :constbval :constval false :location -1 :constval 10 [ 4 0 0 0 0 6 9 8 5 8 2 7 9 8 0 6 9 ] ) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 1 :clause_rels (b 4) :required_rels (b 4) :incompatible_rels (b) :outer_rels (b 4) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies (o 1994 2095) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 4 :varratno 2 :vartype 25 :vartypemod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1) :em_rels (b 4) :em_is_const false :em_is_child false :em_datatype 25 :em_domain (JOINDOMAIN :jd_rels (b 3 4 5 7 8 9 10 11 12)) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 25 :consttypemod -1 :constcollid 100 :constlen -1 :constbval false :constnull false :location -1 :constval 10 [ 4 0 0 0 0 6 9 8 5 8 2 7 9 8 0 6 9 ] ) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 25 :em_domain (JOINDOMAIN :jd_rels (b 3 4 5 7 8 9 10 11 12)) :outer_is_left false :has_hjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) | parallel_safe = true | reason = PROPARELL_SAFE
WARNING: is_parallel_safe: INPUT qual = { (VAR :varno 4 :varratno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1) : (VAR :varno 4 :varratno 2 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 3 :location -1) | parallel_safe = true | reason = PROPARELL_SAFE
WARNING: is_parallel_safe: INPUT qual = { (RESTRICTINFO :clause (OEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args { (VAR :varno 5 :varratno 3 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 3 :location -1) (CONST :consttype 23 :consttypemod -1 :constcollid 0 :constlen 4 :constbval true :constnull false :location -1 :constval 4 [ 1 0 0 0 0 0 0 0 0 0 ] ) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 5) :required_rels (b 5) :incompatible_rels (b) :outer_rels (b 5) :left_rels (b 5) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies (o 1994 2095) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 4 :varratno 2 :vartype 25 :vartypemod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1) :em_rels (b 4) :em_is_const true :em_is_child false :em_datatype 25 :em_domain (JOINDOMAIN :jd_rels (b 3 4 5 7 8 9 10 11 12)) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 23 :consttypemod -1 :constcollid 0 :constlen 4 :constbval true :constnull false :location -1 :constval 4 [ 1 0 0 0 0 0 0 0 0 0 ] ) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_domain (JOINDOMAIN :jd_rels (b 3 4 5 7 8 9 10 11 12)) :outer_is_left false :has_hjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) | parallel_safe = true | reason = PROPARELL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = { (VAR :varno 4 :varratno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1) : (VAR :varno 4 :varratno 2 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 3 :location -1) (CONST :consttype 23 :consttypemod -1 :constcollid 0 :constlen 4 :constbval true :constnull false :location -1 :constval 4 [ 1 0 0 0 0 0 0 0 0 0 ] ) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 5) :required_rels (b 5) :incompatible_rels (b) :outer_rels (b 5) :left_rels (b 5) :right_rels (b) :orclause <> :rinfo_serial 1 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies (o 1994 2095) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 5 :varratno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 1 :location -1) :em_rels (b) :em_is_const true :em_is_child false :em_datatype 23 :em_domain (JOINDOMAIN :jd_rels (b 3 4 5 7 8 9 10 11 12)) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 5 :varratno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 1 :location -1) :em_rels (b 5) :em_is_const false :em_is_child false :em_datatype 23 :em_domain (JOINDOMAIN :jd_rels (b 3 4 5 7 8 9 10 11 12)) :outer_is_left false :has_hjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) | parallel_safe = false | reason = PROPARELL_RESTRICTED
WARNING: is_parallel_safe: INPUT qual = { (RESTRICTINFO :clause (OEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args { (VAR :varno 7 :varratno 6 :vartype 701 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattnosyn 6 :location -1) (CONST :consttype 701 :consttypemod -1 :constcollid 0 :constlen 8 :constbval true :constnull false :location -1 :constval 8 [ 0 0 0 0 0 0 0 0 ] ) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 7) :required_rels (b 7) :incompatible_rels (b) :outer_rels (b 7) :left_rels (b 7) :right_rels (b) :orclause <> :rinfo_serial 2 :eval_cost_startup -1 :eval_cost_per_tuple 0 :norm_select -1 :outer_select -1 :mergeofamilies (o 1994 2095) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 7 :varratno 6 :vartype 701 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattnosyn 6 :location -1) :em_rels (b 7) :em_is_const true :em_is_child false :em_datatype 701 :em_domain (JOINDOMAIN :jd_rels (b 3 4 5 7 8 9 10 11 12)) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 701 :consttypemod -1 :constcollid 0 :constlen 8 :constbval true :constnull false :location -1 :constval 8 [ 0 0 0 0 0 0 0 0 ] ) :em_rels (b 7) :em_is_const true :em_is_child false :em_datatype 701 :em_domain (JOINDOMAIN :jd_rels (b 3 4 5 7 8 9 10 11 12)) :outer_is_left false :has_hjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0) | parallel_safe = true | reason = PROPARELL_SAFE
WARNING: is_parallel_safe: INPUT qual = { (VAR :varno 7 :varratno 4 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattnosyn 4 :location -1) : (VAR :varno 7 :varratno 5 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattnosyn 5 :location -1) : (VAR :varno 7 :varratno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattnosyn 1 :location -1) : (VAR :varno 7 :varratno 3 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattnosyn 3 :location -1) | parallel_safe = true | reason = PROPARELL_SAFE
WARNING: is_parallel_safe: INPUT qual = <<
WARNING: is_parallel_safe: OUTPUT qual = << | parallel_safe = true | reason = PROPARELL_SAFE
WARNING: is_parallel_safe: INPUT qual = { (VAR :varno 8 :varratno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varattnosyn 1 :location -1) : (VAR :varno 8 :varratno 2 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varattnosyn 2 :location -1) : (VAR :varno 8 :varratno 3 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 8 :varattnosyn 3 :location -1) | parallel_safe = true | reason = PROPARELL_SAFE
WARNING: is_parallel_safe: INPUT qual = { (RESTRICTINFO :clause (OEXPR :opno 98 :opfuncid 67 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args { (VAR :varno 9 :varratno 2 :vartype 25 :vartypemod -1 :varcollid
```


[illegible]

```

d-1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 3 :location -1} {VAR :varno 4 :varrattno 1 :varvtype 23 :varvtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 4) :required_relds (b 3 4) :incompatible_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b 4) :orclause <> :rinfo_s
erial 25 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varrattno 3 :varvtype 23 :varvtypm
od -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 3 :location -1} :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOIN
DOMAIN :jd_relds (b 3 4 5 7 8 9 10 11 12))) :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varrattno 1 :varvtype 23 :varvtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0
:varnosyn 4 :varrattnosyn 1 :location -1} :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 3 4 5 7 8 9 10 11 12))) :outer_is_left true :hashjo
inoperator 96 :left_bucketsize 0.1 :right_bucketsize 0.1 :left_mcvfreq 0 :right_mcvfreq 0 :left_hasheqoperator 96 :right_hasheqoperator 96}
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varrattno 3 :varvtype 23 :va
rtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 3 :location -1} {VAR :varno 4 :varrattno 1 :varvtype 23 :varvtypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 4) :required_relds (b 3 4) :incompatible_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b 4) :orclause <> :rinfo_s
erial 25 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varrattno 3 :varvtype 23 :varvtypm
od -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 3 :location -1} :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOI
NDOMAIN :jd_relds (b 3 4 5 7 8 9 10 11 12))) :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varrattno 1 :varvtype 23 :varvtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype
0 :varnosyn 4 :varrattnosyn 1 :location -1} :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 3 4 5 7 8 9 10 11 12))) :outer_is_left true :hashj
oinoperator 96 :left_bucketsize 0.1 :right_bucketsize 0.1 :left_mcvfreq 0 :right_mcvfreq 0 :left_hasheqoperator 96 :right_hasheqoperator 96} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 7 :varrattno 1 :varvtype 23 :varvtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varrattnosyn 1 :location -1}
{VAR :varno 3 :varrattno 1 :varvtype 23 :varvtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 1 :location -1} | parallel_safe = true | reason = PRO
PARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 3 :varrattno 3 :varvtype 23 :va
rtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 3 :location -1} {VAR :varno 4 :varrattno 1 :varvtype 23 :varvtypmod -1 :varcollid 0 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varrattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 3 4) :required_relds (b 3 4) :incompatible_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b 4) :orclause <> :rinfo_s
erial 25 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selec 0.005 :outer_selec -1 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 3 :varrattno 3 :varvtype 23 :varvtypm
od -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 3 :location -1} :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOI
NDOMAIN :jd_relds (b 3 4 5 7 8 9 10 11 12))) :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 4 :varrattno 1 :varvtype 23 :varvtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype
0 :varnosyn 4 :varrattnosyn 1 :location -1} :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 3 4 5 7 8 9 10 11 12))) :outer_is_left true :hashj
oinoperator 96 :left_bucketsize 0.1 :right_bucketsize 0.1 :left_mcvfreq 0 :right_mcvfreq 0 :left_hasheqoperator 96 :right_hasheqoperator 96} | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 10 :varrattno 1 :varvtype 23 :varvtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 10 :varrattnosyn 1 :location -
1} {VAR :varno 7 :varrattno 1 :varvtype 23 :varvtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varrattnosyn 1 :location -1} | parallel_safe = true | reason = PR
OPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((AGGREG :aggfnoid 2136 :aggtype 701 :aggcollid 0 :inputcollid 0 :aggtranstype 701 :aggargtypes (o 701) :aggdirectargs <> :args ((TARGETENTRY :expr {VAR :varno 5
:varrattno 4 :varvtype 701 :varvtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varrattnosyn 4 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl
0 :resorigcol 0 :resjunk false)} :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1
))
WARNING: is_parallel_safe: OUTPUT qual = ((AGGREG :aggfnoid 2136 :aggtype 701 :aggcollid 0 :inputcollid 0 :aggtranstype 701 :aggargtypes (o 701) :aggdirectargs <> :args ((TARGETENTRY :expr {VAR :varno 5
:varrattno 4 :varvtype 701 :varvtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varrattnosyn 4 :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl
0 :resorigcol 0 :resjunk false)} :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -
1))
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 5 :varrattno 4 :varvtype 701 :varvtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varrattnosyn 4 :location -1
})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 5 :varrattno 4 :varvtype 701 :varvtypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varrattnosyn 4 :location -
1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: { QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowRTE false
:hasGroupRTE false
:isReturn false
:cteList <
:rttable (
:RANGETBENTRY
:alias <
:eref
:ALIAS
:aliasname part
:colnames ("p_partkey" "p_name" "p_mfgr" "p_brand" "p_type" "p_size"
"p_container" "p_retailprice" "p_comment")
}
:rtkind 0
:reld 16551
:inh false
:relink r
:rellockmode 1
:perminfoindex 1
:tablesample <
:lateral false
:inFromCl true
:securityQuals <
}
:RANGETBENTRY
:alias <
:eref
:ALIAS
:aliasname supplier_secure
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
:rtkind 1
:subquery <
:security_barrier false
:reld 17511
:inh false
:relink v
:rellockmode 1
:perminfoindex 2
:lateral false
:inFromCl true
:securityQuals <
}
:RANGETBENTRY
:alias <
:eref
:ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtkind 0
:reld 16536
:inh false
:relink r
:rellockmode 1

```

```

:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname region
:colnames ("r_regionkey" "r_name" "r_comment")
}
:rtekind 0
:relid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 6
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname n
:colnames <>
}
}
:eref
{ALIAS
:aliasname n
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
}
:rtekind 0
:relid 16536
:inh false
:relkind r
:rellockmode 1
:perminfoindex 7
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname r
:colnames <>
}
}
:eref
{ALIAS
:aliasname r
:colnames ("r_regionkey" "r_name" "r_comment")
}
}
:rtekind 0
:relid 16531
:inh false
:relkind r
:rellockmode 1
:perminfoindex 8
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_supkey" "ps_availqty" "ps_supplycost" "p
s_comment")
}
}
:rtekind 0
:relid 16556
:inh false
:relkind r
:rellockmode 1
:perminfoindex 9
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
}
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 17511
:inh false
:relkind v
:rellockmode 1
:perminfoindex 10
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "

```

```

        s_acctbal" "s_comment")
    }
    :rtekind 0
    :relid 16541
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 11
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    {RANGETBLENTTRY
    :alias
    {ALIAS
    :aliasname n
    :colnames <>
    }
    :eref
    {ALIAS
    :aliasname n
    :colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
    }
    :rtekind 0
    :relid 16536
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 12
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    {RANGETBLENTTRY
    :alias
    {ALIAS
    :aliasname r
    :colnames <>
    }
    :eref
    {ALIAS
    :aliasname r
    :colnames ("r_regionkey" "r_name" "r_comment")
    }
    :rtekind 0
    :relid 16531
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 13
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    }
    :rteperminfos (
    {RTEPERMISSIONINFO
    :relid 16551
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 0 10 12 13)
    :insertedCols (b)
    :updatedCols (b)
    }
    {RTEPERMISSIONINFO
    :relid 17511
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10 11 12 13 14)
    :insertedCols (b)
    :updatedCols (b)
    }
    {RTEPERMISSIONINFO
    :relid 17515
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 11)
    :insertedCols (b)
    :updatedCols (b)
    }
    {RTEPERMISSIONINFO
    :relid 16536
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10)
    :insertedCols (b)
    :updatedCols (b)
    }
    {RTEPERMISSIONINFO
    :relid 16531
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
    }
    {RTEPERMISSIONINFO
    :relid 16541
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 9 10 11 12 13 14)
    :insertedCols (b)
    :updatedCols (b)
    }
    {RTEPERMISSIONINFO
    :relid 16536
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 10)
    :insertedCols (b)
    :updatedCols (b)
    }
    {RTEPERMISSIONINFO
    :relid 16531
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 9)
    :insertedCols (b)
    :updatedCols (b)
    }
    {RTEPERMISSIONINFO
    :relid 16556
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 9 10 11 12)
    :insertedCols (b)
    :updatedCols (b)
    }
    {RTEPERMISSIONINFO
    :relid 17511
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
    }
    )

```

```

}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16536
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16531
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 6
}
)
:quals (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 6
:location -1
}
){CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:rarg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 7
}
{RANGETBLREF
:rtindex 8
}
)
:quals <>
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 1
:location -1
}
{VAR
:varno 6
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 4
:location -1
}
)
:location -1
}
)
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0

```

```

:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 3
:location -1
}
{VAR
:varno 8
:varattno 1
:varatype 23
:varatypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 1
:location -1
}
}
:location -1
}
{OPEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 8
:varattno 2
:varatype 25
:varatypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 8
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 9
}
)
:quals (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 9
:varattno 3
:varatype 23
:varatypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 3
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 100 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
}
:larg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 11
}
)
:quals (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 11
:varattno 6
:varatype 701
:varatypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 11
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false

```

```

        :location -1
        :constvalue 8 [ 0 0 0 0 0 0 0 0 ]
    }
}
:location -1
}
}
}
:arg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 12
}
{RANGETBLREF
:rtindex 13
}
)
:quals <>
}
:usingClause <>
:join_using_alias <>
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 12
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 12
:varattnosyn 1
:location -1
}
{VAR
:varno 11
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 11
:varattnosyn 4
:location -1
}
)
:location -1
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 12
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 12
:varattnosyn 3
:location -1
}
{VAR
:varno 13
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 13
:varattnosyn 1
:location -1
}
)
:location -1
}
{OEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 13
:varattno 2
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 13
:varattnosyn 2
:location -1
}
{CONST
:consttype 25
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
:usingClause <>
:join_using_alias <>
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 11

```

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:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 11
:varattnosyn 1
:location -1
}
{VAR
:varno 9
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 2
:location -1
}
}
:location -1
}
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 11
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 11
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:alias <>
:rtindex 0
}
{RANGETBLREF
:rtindex 4
}
{RANGETBLREF
:rtindex 5
}
)
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 297
}
{VAR
:varno 9
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 1
:location -1
}
}
:location 307
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 1
:location -1
}
{VAR
:varno 9
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 9
:varattnosyn 2
:location -1
}
}
:location 342
}
{OPEXPR

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:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 6
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location 367
  }
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 376
    :constvalue 4 [ 15 0 0 0 0 0 0 0 ]
  }
)
:location 374
}
{OEXPR
:opno 1209
:opfuncid 850
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
  {VAR
    :varno 1
    :varattno 5
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 391
  }
  {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 403
    :constvalue 10 [ 40 0 0 0 37 66 82 65 83 83 ]
  }
)
:location 398
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 6
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 6
    :varattnosyn 4
    :location -1
  }
  {VAR
    :varno 4
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 1
    :location 438
  }
)
:location 436
}
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 4
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 3
    :location 462
  }
  {VAR
    :varno 5
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 5
    :varattnosyn 1
    :location 476
  }
)
:location 474
}
{OEXPR
:opno 98
:opfuncid 67
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100

```

```

:args (
  {VAR
    :varno 5
    :varattno 2
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 5
    :varattnosyn 2
    :location 507
  }
  {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 523
    :constvalue 10 [ 40 0 0 0 69 85 82 79 80 69 ]
  }
)
:location 521
}
{OPEXPR
  :opno 670
  :opfuncid 293
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
      :varno 9
      :varattno 4
      :vartype 701
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 9
      :varattnosyn 4
      :location -1
    }
    {SUBPLAN
      :sublinktype 4
      :testexpr <>
      :paramids <>
      :plan_id 1
      :plan_name SubPlan\ 1
      :firstColType 701
      :firstColTypmod -1
      :firstColCollation 0
      :useHashTable false
      :unknownEqFalse false
      :parallel_safe false
      :setParam <>
      :parParam (i 0)
      :args (
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location 840
        }
      )
    }
  )
  :startup_cost 0
  :per_call_cost 34072.729499999994
}
:location 558
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 6
      :varattno 6
      :vartype 701
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 6
      :varattnosyn 6
      :location -1
    }
  }
  :resno 1
  :resname s_acctbal
  :ressortgroupref 1
  :resorigtbl 17511
  :resorigcol 6
  :resjunk false
}
{TARGETENTRY
  :expr
  {VAR
    :varno 6
    :varattno 2
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 6
    :varattnosyn 2
    :location -1
  }
}
:resno 2
:resname s_name
:ressortgroupref 3
:resorigtbl 17511
:resorigcol 2
:resjunk false
}
{TARGETENTRY
  :expr
  {VAR
    :varno 4
    :varattno 2
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 2
    :location 86
  }
}

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```

:resno 3
:resname n_name
:ressortgroupref 2
:resorigtbl 16536
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 102
}
:resno 4
:resname p_partkey
:ressortgroupref 4
:resorigtbl 16551
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 121
}
:resno 5
:resname p_mfgr
:ressortgroupref 0
:resorigtbl 16551
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 6
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 3
:location -1
}
:resno 6
:resname s_address
:ressortgroupref 0
:resorigtbl 17511
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 6
:varattno 5
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 5
:location -1
}
:resno 7
:resname s_phone
:ressortgroupref 0
:resorigtbl 17511
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 6
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 7
:location -1
}
:resno 8
:resname s_comment
:ressortgroupref 0
:resorigtbl 17511
:resorigcol 7
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 670
:sortop 674
:reverse_sort true
:nulls_first true
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 2
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
{SORTGROUPCLAUSE
:tleSortGroupRef 3

```

```

:equip 98
:sorttop 664
:reverse_sort false
:nulls_first false
:hashable true
}

(SORTGROUPCLAUSE
:tleSortGroupRef 4
:equip 96
:sorttop 97
:reverse_sort false
:nulls_first false
:hashable true
}

)

:limitOffset 0
:limitCount
(CONST
:consttype 20
:consttypmod -1
:constcollid 0
:constlen 8
:constbval true
:constnull false
:constnull false
:location -1 :constval 8 [ 100 0 0 0 0 0 0 0 ]
)

:limitOption 0
:rowMarks 0
:setOperations
:constraintDeps <
:withCheckOptions <
:stmt_location 0
:stmt_len 0
)

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 1289 :opfuncid 850 :presulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 1 :varattno 5 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1) (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbval true :is_parallel_safe :location -1 :constval 10 [ 40 0 0 0 37 66 82 65 83 83 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 15 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :right_hashoperator 0 :right_hashoperator 0) (RESTRICTINFO :clause (OEXPR :opno 96 :opfuncid 65 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 6 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) (CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbval true :constnull false :location -1 :constval 4 [ 15 0 0 0 0 0 0 0 ])) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbval true :constnull false :location -1 :constval 4 [ 15 0 0 0 0 0 0 0 ])) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0))
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 1289 :opfuncid 850 :presulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ((VAR :varno 1 :varattno 5 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1) (CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbval true :is_parallel_safe :location -1 :constval 10 [ 40 0 0 0 37 66 82 65 83 83 ])) :location -1) :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 15 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :right_hashoperator 0 :right_hashoperator 0) (RESTRICTINFO :clause (OEXPR :opno 96 :opfuncid 65 :presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 6 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) (CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbval true :constnull false :location -1 :constval 4 [ 15 0 0 0 0 0 0 0 ])) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :right_em (EQUIVALENCEMEMBER :em_expr (CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbval true :constnull false :location -1 :constval 4 [ 15 0 0 0 0 0 0 0 ])) :em_relds (b) :em_is_const true :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0))
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1)) (VAR :varno 1 :varattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1)) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 1 :varattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1)) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 4 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 1 :location -1) (VAR :varno 4 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 3 :location -1)) (VAR :varno 4 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 3 :location -1)) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 4 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 2 :location -1) (VAR :varno 4 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 4 :location -1)) (VAR :varno 4 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattnosyn 4 :location -1)) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 5 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 1 :location -1) (VAR :varno 5 :varattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 3 :location -1)) (VAR :varno 5 :varattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 3 :location -1)) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 5 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 2 :location -1) (VAR :varno 5 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 4 :location -1)) (VAR :varno 5 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattnosyn 4 :location -1)) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 6 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 1 :location -1) (VAR :varno 6 :varattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 3 :location -1)) (VAR :varno 6 :varattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 3 :location -1)) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 6 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 2 :location -1) (VAR :varno 6 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 4 :location -1)) (VAR :varno 6 :varattno 4 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 4 :location -1)) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 7 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattnosyn 1 :location -1) (VAR :varno 7 :varattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattnosyn 3 :location -1)) (VAR :varno 7 :varattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattnosyn 3 :location -1)) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 7 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturning
```

[illegible]

[illegible]

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25 :var1pmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 7 :varattnosyn 1 :location -17)
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 6 :varattno 6 :vartype 701 :varcollid -1 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 6 :location -1} {VAR :varno 2 :varattno 2 :vartype 25 :var1pmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 2 :location -1} {VAR :varno 6 :varattno 3 :vartype 2
```

```
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 96 :opfncund 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 6 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 1 :location -1) (VAR :varno 11 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 11 :varattnosyn 3 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_rels (b 6 11) :required_rels (b 6 11) :incompatible_rels (b) :outer_rels (b) :left_rels (b 6) :right_rels (b 11) :orclause <- 1 :info_serial 28 :eval_cost.startup 0 :eval_cost.per_tupletuple 0.0025 :norm_select 8.449514152936207e-05 :outer_select -1 :mergeofamilies (o 1976) :left_em {EQUIVALENCEMEMBER :em_expr (VAR :varno 6 :varattno 1
```

[illegible]

```

1  is_parallel_safe: INPUT qual = { (VAR :varno 4 :varattno 2 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 2 :location -1)
2  (VAR :varno 9 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 1 :location -1) (VAR :varno 9 :varattno 4 :vartype 701 :
3  vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 4 :location -1) (VAR :varno 11 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrel
4  s (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 11 :varattsyns 1 :location -1) (VAR :varno 7 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0
5  :varnosyn 7 :varattsyns 1 :location -1) (VAR :varno 5 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattsyns 1 :location -1)
6  }
7  WARNING: is_parallel_safe: OUTPUT qual = { (VAR :varno 9 :varattno 2 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 2 :location -1)
8  (VAR :varno 9 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 1 :location -1) (VAR :varno 9 :varattno 4 :vartype 701 :
9  vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 4 :location -1) (VAR :varno 11 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrel
10 ls (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 11 :varattsyns 1 :location -1) (VAR :varno 7 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype
11 0 :varnosyn 7 :varattsyns 1 :location -1) (VAR :varno 5 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattsyns 1 :location -1)
12  }
13  | parallel_safe = true | reason = PROPAPARALLEL_SAFE
14  WARNING: is_parallel_safe: INPUT qual = { (PROPARALLEL_SAFE :clause (OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcolld 0 :inputcolld 0 :args ((VAR :varno 4 :varattno 3 :vartype 23 :va
15  rtypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattsyns 3 :location -1) (VAR :varno 5 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (
16 b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattsyns 1 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
17 :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 4 5) :required_relds (b 4 5) :incompatible_relds (b) :outer_relds (b) :left_relds (b 4) :right_relds (b 5) :orclause <> :rinfo: s
18 erial 31 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.005 :outer_selc -1 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 4 :varattno 3 :vartype 23 :vartypmo
19 d -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattsyns 3 :location -1) :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOIN
20 DOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 5 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype
21 0 :varnosyn 5 :varattsyns 1 :location -1) :em_relds (b 5) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :outer_is_left true :ha
22 shjoinoperator 96 :left_bucketsize 0.1 :right_bucketsize 0.1 :left_mcvfreq 0 :right_mcvfreq 0 :left_hashoperator 96 :right_hashoperator 96))
23 WARNING: is_parallel_safe: OUTPUT qual = { (RESTRICTINFO :clause (OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcolld 0 :inputcolld 0 :args ((VAR :varno 4 :varattno 3 :vartype 23 :v
24 artypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattsyns 3 :location -1) (VAR :varno 5 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels
25 (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 5 :varattsyns 1 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
26 :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 4 5) :required_relds (b 4 5) :incompatible_relds (b) :outer_relds (b) :left_relds (b 4) :right_relds (b 5) :orclause <> :rinfo: s
27 erial 31 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.005 :outer_selc -1 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 4 :varattno 3 :vartype 23 :vartypmo
28 d -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattsyns 3 :location -1) :em_relds (b 4) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOIN
29 DOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 5 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype
30 0 :varnosyn 5 :varattsyns 1 :location -1) :em_relds (b 5) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :outer_is_left true :ha
31 shjoinoperator 96 :left_bucketsize 0.1 :right_bucketsize 0.1 :left_mcvfreq 0 :right_mcvfreq 0 :left_hashoperator 96 :right_hashoperator 96))
32 WARNING: is_parallel_safe: INPUT qual = { (VAR :varno 4 :varattno 6 :vartype 701 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattsyns 6 :location -1)
33 (VAR :varno 6 :varattno 6 :vartype 701 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattsyns 6 :location -1)
34 | parallel_safe = true | reason = PROPAPARALLEL_SAFE
35 WARNING: is_parallel_safe: OUTPUT qual = { (VAR :varno 4 :varattno 6 :vartype 701 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattsyns 6 :location -1)
36 (VAR :varno 6 :varattno 6 :vartype 701 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattsyns 6 :location -1)
37 }
38 WARNING: is_parallel_safe: INPUT qual = { (VAR :varno 4 :varattno 2 :vartype 25 :vartypmod -1 :varcolld 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 4 :varattsyns 2 :location -1)
39 | parallel_safe = true | reason = PROPAPARALLEL_SAFE
40 WARNING: is_parallel_safe: INPUT qual = { (VAR :varno 6 :varattno 2 :vartype 25 :vartypmod -1 :varcolld 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattsyns 2 :location -1)
41 | parallel_safe = true | reason = PROPAPARALLEL_SAFE
42 WARNING: is_parallel_safe: OUTPUT qual = { (VAR :varno 6 :varattno 2 :vartype 25 :vartypmod -1 :varcolld 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattsyns 2 :location
43 -1) | parallel_safe = true | reason = PROPAPARALLEL_SAFE
44 WARNING: is_parallel_safe: INPUT qual = { (VAR :varno 9 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 1 :location -1)
45 | parallel_safe = true | reason = PROPAPARALLEL_SAFE
46 WARNING: is_parallel_safe: OUTPUT qual = { (VAR :varno 9 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 1 :location -1)
47 | parallel_safe = true | reason = PROPAPARALLEL_SAFE
48 WARNING: is_parallel_safe: INPUT qual = { (VAR :varno 6 :varattno 6 :vartype 701 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattsyns 6 :location -1)
49 (VAR :varno 6 :varattno 6 :vartype 701 :vartypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattsyns 6 :location -1)
50 | parallel_safe = true | reason = PROPAPARALLEL_SAFE
51 WARNING: is_parallel_safe: INPUT qual = { (RESTRICTINFO :clause (OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcolld 0 :inputcolld 0 :args ((VAR :varno 1 :varattno 1 :vartype 23 :va
52 rtypmod -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattsyns 1 :location -1) (VAR :varno 9 :varattno 1 :vartype 23 :vartypmod -1 :varcolld 0 :varnullingrels (
53 b) :varlevelsup 0 :varreturningtype 0 :varnosyn 9 :varattsyns 1 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
54 :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 9) :required_relds (b 1 9) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b 9) :orclause <> :rinfo: s
55 erial 26 :eval_cost.startup 0 :eval_cost.per_tuple 0.0025 :norm_selc 0.005 :outer_selc -1 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 1 :vartype 23 :vartypmo
56 d -1 :varcolld 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattsyns 1 :location -1) :em_relds (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOIN
57 DOMAIN :jd_relds (b 1 4 5 6 7 8 9 11 12 13))) :right_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 9 :varattno 1 :vartype 23 :vartypmod -1 :varcol
```

[illegible]

WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 6 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 6 :location -1} (VAR :varno 6 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 2 :location -1} (VAR :varno 4 :varattno 2 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} (VAR :varno 1 :varattno 3 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} (VAR :varno 6 :varattno 5 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 5 :location -1} (VAR :varno 6 :varattno 7 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 6 :varattnosyn 7 :location -1})) | parallel_safe = true | reason = PROPARALLEL_SAFE

QUERY PLAN

Limit (cost=6406085.83..6406085.84 rows=1 width=182) (actual time=584607.424..584610.744 rows=100.00 loops=1)
Output: s.s_acctbal, s.s_name, nation.n_name, part.p_partkey, part.p_mfgr, s.s_address, s.s_phone, s.s_comment
Buffers: shared hit=2597844 read=63188015
-> Sort (cost=6406085.83..6406085.84 rows=1 width=182) (actual time=584607.421..584610.734 rows=100.00 loops=1)
Output: s.s_acctbal, s.s_name, nation.n_name, part.p_partkey, part.p_mfgr, s.s_address, s.s_phone, s.s_comment
Sort Key: s.s_acctbal DESC, nation.n_name, s.s_name, part.p_partkey
Sort Method: top-N heapsort Memory: 68kB
Buffers: shared hit=2597844 read=63188015
-> Hash Join (cost=8740.25..6406085.82 rows=1 width=182) (actual time=87800.529..584596.173 rows=2440.00 loops=1)
Output: s.s_acctbal, s.s_name, nation.n_name, part.p_partkey, part.p_mfgr, s.s_address, s.s_phone, s.s_comment
Hash Cond: ((ps.ps_partkey = part.p_partkey) AND (ps.ps_supplycost = (SubPlan 1)))
Buffers: shared hit=2597835 read=63188015
-> Gather (cost=2059.64..27801.98 rows=374 width=175) (actual time=27.280..163.995 rows=748884.00 loops=1)
Output: s.s_acctbal, s.s_name, s.s_address, s.s_phone, s.s_comment, ps.ps_partkey, ps.ps_supplycost, nation.n_name
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=1841 read=19509
-> Hash Join (cost=1059.64..26764.58 rows=156 width=175) (actual time=16.452..192.346 rows=249628.00 loops=3)
Output: s.s_acctbal, s.s_name, s.s_address, s.s_phone, s.s_comment, ps.ps_partkey, ps.ps_supplycost, nation.n_name
Hash Cond: (s_1.s_suppley = s.s_suppley)
Buffers: shared hit=1841 read=19509
Worker 0: actual time=11.502..236.187 rows=329816.00 loops=1
Buffers: shared hit=626 read=8628
Worker 1: actual time=11.614..238.557 rows=334052.00 loops=1
Buffers: shared hit=625 read=8758
-> Hash Semi Join (cost=527.44..26131.78 rows=26410 width=20) (actual time=9.010..101.495 rows=52638.67 loops=3)
Output: ps.ps_suppley, ps.ps_partkey, ps.ps_supplycost, s_1.s_suppley
Hash Cond: (ps.ps_suppley = s_1.s_suppley)
Buffers: shared hit=1055 read=19509
Worker 0: actual time=4.135..125.940 rows=69663.00 loops=1
Buffers: shared hit=364 read=8628
Worker 1: actual time=4.323..125.760 rows=70754.00 loops=1
Buffers: shared hit=363 read=8758
-> Parallel Seq Scan on public.partsupp ps (cost=0.00..24358.70 rows=362603 width=16) (actual time=1.018..52.241 rows=290354.00 loops=3)
Output: ps.ps_partkey, ps.ps_suppley, ps.ps_avallqty, ps.ps_supplycost, ps.ps_comment
Filter: (ps.ps_avallqty > 100)
Rows Removed by Filter: 2979
Buffers: shared hit=275 read=19509
Worker 0: actual time=0.785..67.428 rows=384344.00 loops=1
Buffers: shared hit=104 read=8628
Worker 1: actual time=0.867..69.160 rows=390657.00 loops=1
Buffers: shared hit=103 read=8758
-> Hash (cost=516.66..516.66 rows=862 width=4) (actual time=7.975..8.526 rows=1994.00 loops=3)
Output: s_1.s_suppley
Buckets: 2048 (originally 1024) Batches: 1 (originally 1) Memory Usage: 87kB
Buffers: shared hit=780
Worker 0: actual time=3.338..4.096 rows=1994.00 loops=1
Buffers: shared hit=260
Worker 1: actual time=3.441..4.200 rows=1994.00 loops=1
Buffers: shared hit=260
-> Hash Semi Join (cost=42.17..516.66 rows=862 width=4) (actual time=0.202..8.096 rows=1994.00 loops=3)
Output: s_1.s_suppley
Hash Cond: (s_1.s_nationkey = n_1.n_nationkey)
Buffers: shared hit=780
Worker 0: actual time=0.218..3.802 rows=1994.00 loops=1
Buffers: shared hit=260
Worker 1: actual time=0.243..3.914 rows=1994.00 loops=1
Buffers: shared hit=260
-> Seq Scan on public.supplier s_1 (cost=0.00..435.52 rows=10780 width=8) (actual time=0.052..6.341 rows=10026.00 loops=3)
Output: s_1.s_name, s_1.s_address, s_1.s_nationkey, s_1.s_phone, s_1.s_acctbal, s_1.s_comment
Filter: ((s_1.s_acctbal > '0':double precision) AND (s_1.s_acctbal > '0':double precision))
Rows Removed by Filter: 974
Buffers: shared hit=774
Worker 0: actual time=0.060..2.018 rows=10026.00 loops=1
Buffers: shared hit=258
Worker 1: actual time=0.065..2.094 rows=10026.00 loops=1
Buffers: shared hit=258
-> Hash (cost=41.97..41.97 rows=16 width=4) (actual time=0.138..0.686 rows=20.00 loops=3)
Output: n_1.n_nationkey
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=6
Worker 0: actual time=0.147..0.902 rows=20.00 loops=1
Buffers: shared hit=2
Worker 1: actual time=0.164..0.920 rows=20.00 loops=1
Buffers: shared hit=2
-> Hash Join (cost=20.68..41.97 rows=16 width=4) (actual time=0.115..0.675 rows=20.00 loops=3)
Output: n_1.n_nationkey
Hash Cond: (n_1.n_regionkey = r_1.r_regionkey)
Buffers: shared hit=6
Worker 0: actual time=0.130..0.894 rows=20.00 loops=1
Buffers: shared hit=2
Worker 1: actual time=0.147..0.912 rows=20.00 loops=1
Buffers: shared hit=2
-> Seq Scan on public.nation n_1 (cost=0.00..18.10 rows=810 width=8) (actual time=0.039..0.043 rows=50.00 loops=3)
Output: n_1.n_nationkey, n_1.n_name, n_1.n_regionkey, n_1.n_comment
Buffers: shared hit=3
Worker 0: actual time=0.045..0.048 rows=50.00 loops=1
Buffers: shared hit=1
Worker 1: actual time=0.051..0.054 rows=50.00 loops=1
Buffers: shared hit=1
-> Hash (cost=20.62..20.62 rows=4 width=4) (actual time=0.064..0.568 rows=2.00 loops=3)
Output: r_1.r_regionkey
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=3
Worker 0: actual time=0.076..0.831 rows=2.00 loops=1
Buffers: shared hit=1
Worker 1: actual time=0.087..0.842 rows=2.00 loops=1
Buffers: shared hit=1
-> Seq Scan on public.region r_1 (cost=0.00..20.62 rows=4 width=4) (actual time=0.047..0.552 rows=2.00 loops=3)
Output: r_1.r_regionkey
Filter: (r_1.r_name = 'EUROPE':text)
Rows Removed by Filter: 8
Buffers: shared hit=3
Worker 0: actual time=0.059..0.815 rows=2.00 loops=1
Buffers: shared hit=1
Worker 1: actual time=0.070..0.826 rows=2.00 loops=1
Buffers: shared hit=1
-> Hash (cost=531.33..531.33 rows=70 width=167) (actual time=7.428..7.907 rows=7976.00 loops=3)
Output: s.s_acctbal, s.s_name, s.s_address, s.s_phone, s.s_comment, s.s_suppley, nation.n_name
Buckets: 8192 (originally 1024) Batches: 1 (originally 1) Memory Usage: 1458kB
Buffers: shared hit=786
Worker 0: actual time=7.351..8.392 rows=7976.00 loops=1
Buffers: shared hit=262
Worker 1: actual time=7.280..7.653 rows=7976.00 loops=1
Buffers: shared hit=262
-> Hash Join (cost=84.43..531.33 rows=70 width=167) (actual time=0.132..4.507 rows=7976.00 loops=3)
Output: s.s_acctbal, s.s_name, s.s_address, s.s_phone, s.s_comment, s.s_suppley, nation.n_name
Hash Cond: (s.s_nationkey = n.n_nationkey)
Buffers: shared hit=786
Worker 0: actual time=0.132..4.977 rows=7976.00 loops=1
Buffers: shared hit=262
Worker 1: actual time=0.129..4.281 rows=7976.00 loops=1
Buffers: shared hit=262
-> Seq Scan on public.supplier s (cost=0.00..405.94 rows=10780 width=139) (actual time=0.007..1.887 rows=10026.00 loops=3)
Output: s.s_suppley, s.s_name, s.s_address, s.s_nationkey, s.s_phone, s.s_acctbal, s.s_comment
Filter: ((s.s_acctbal > '0':double precision))
Rows Removed by Filter: 974
Buffers: shared hit=774
Worker 0: actual time=0.008..1.820 rows=10026.00 loops=1
Buffers: shared hit=258
Worker 1: actual time=0.005..1.774 rows=10026.00 loops=1
Buffers: shared hit=258
-> Hash (cost=84.41..84.41 rows=1 width=40) (actual time=0.114..0.589 rows=20.00 loops=3)
Output: nation.n_name, nation.n_nationkey, n.n_nationkey
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=12
Worker 0: actual time=0.115..1.148 rows=20.00 loops=1
Buffers: shared hit=4
Worker 1: actual time=0.115..0.485 rows=20.00 loops=1

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    Buffers: shared hit=4
-> Hash Join (cost=63.05..84.41 rows=1 width=40) (actual time=0.089..0.578 rows=20.00 loops=3)
    Output: nation.n_name, nation.n_nationkey, n.n_nationkey
    Hash Cond: (nation.n_nationkey = n.n_nationkey)
    Buffers: shared hit=12
    Worker 0: actual time=0.089..1.138 rows=20.00 loops=1
        Buffers: shared hit=4
    Worker 1: actual time=0.089..0.474 rows=20.00 loops=1
        Buffers: shared hit=4
-> Hash Join (cost=20.68..41.97 rows=16 width=36) (actual time=0.029..0.049 rows=20.00 loops=3)
    Output: nation.n_name, nation.n_nationkey
    Hash Cond: (nation.n_regionkey = region.r_regionkey)
    Buffers: shared hit=6
    Worker 0: actual time=0.032..0.049 rows=20.00 loops=1
        Buffers: shared hit=2
    Worker 1: actual time=0.031..0.047 rows=20.00 loops=1
        Buffers: shared hit=2
-> Seq Scan on public.nation (cost=0.00..18.10 rows=810 width=40) (actual time=0.003..0.006 rows=50.00 loops=3)
    Output: nation.n_nationkey, nation.n_name, nation.n_regionkey, nation.n_comment
    Buffers: shared hit=3
    Worker 0: actual time=0.003..0.007 rows=50.00 loops=1
        Buffers: shared hit=1
    Worker 1: actual time=0.003..0.006 rows=50.00 loops=1
        Buffers: shared hit=1
-> Hash (cost=20.62..20.62 rows=4 width=4) (actual time=0.015..0.019 rows=2.00 loops=3)
    Output: region.r_regionkey
    Buckets: 1024 Batches: 1 Memory Usage: 9kB
    Buffers: shared hit=3
    Worker 0: actual time=0.015..0.015 rows=2.00 loops=1
        Buffers: shared hit=1
    Worker 1: actual time=0.015..0.016 rows=2.00 loops=1
        Buffers: shared hit=1
-> Seq Scan on public.region (cost=0.00..20.62 rows=4 width=4) (actual time=0.006..0.008 rows=2.00 loops=3)
    Output: region.r_regionkey
    Filter: (region.r_name = 'EUROPE'::text)
    Rows Removed by Filter: 8
    Buffers: shared hit=3
    Worker 0: actual time=0.007..0.008 rows=2.00 loops=1
        Buffers: shared hit=1
    Worker 1: actual time=0.008..0.009 rows=2.00 loops=1
        Buffers: shared hit=1
-> Hash (cost=42.17..42.17 rows=16 width=4) (actual time=0.051..0.288 rows=5.00 loops=3)
    Output: n.n_nationkey
    Buckets: 1024 Batches: 1 Memory Usage: 9kB
    Buffers: shared hit=6
    Worker 0: actual time=0.050..0.740 rows=5.00 loops=1
        Buffers: shared hit=2
    Worker 1: actual time=0.051..0.068 rows=5.00 loops=1
        Buffers: shared hit=2
-> HashAggregate (cost=42.01..42.17 rows=16 width=4) (actual time=0.044..0.279 rows=5.00 loops=3)
    Output: n.n_nationkey
    Group Key: n.n_nationkey
    Batches: 1 Memory Usage: 32kB
    Buffers: shared hit=6
    Worker 0: actual time=0.042..0.731 rows=5.00 loops=1
        Batches: 1 Memory Usage: 32kB
        Buffers: shared hit=2
    Worker 1: actual time=0.043..0.059 rows=5.00 loops=1
        Batches: 1 Memory Usage: 32kB
        Buffers: shared hit=2
-> Hash Join (cost=20.68..41.97 rows=16 width=4) (actual time=0.025..0.270 rows=20.00 loops=3)
    Output: n.n_nationkey
    Hash Cond: (n.n_regionkey = r.r_regionkey)
    Buffers: shared hit=6
    Worker 0: actual time=0.023..0.721 rows=20.00 loops=1
        Buffers: shared hit=2
    Worker 1: actual time=0.025..0.050 rows=20.00 loops=1
        Buffers: shared hit=2
-> Seq Scan on public.nation n (cost=0.00..18.10 rows=810 width=8) (actual time=0.003..0.006 rows=50.00 loops=3)
    Output: n.n_nationkey, n.n_name, n.n_regionkey, n.n_comment
    Buffers: shared hit=3
    Worker 0: actual time=0.003..0.006 rows=50.00 loops=1
        Buffers: shared hit=1
    Worker 1: actual time=0.003..0.006 rows=50.00 loops=1
        Buffers: shared hit=1
-> Hash (cost=20.62..20.62 rows=4 width=4) (actual time=0.012..0.246 rows=2.00 loops=3)
    Output: r.r_regionkey
    Buckets: 1024 Batches: 1 Memory Usage: 9kB
    Buffers: shared hit=3
    Worker 0: actual time=0.010..0.696 rows=2.00 loops=1
        Buffers: shared hit=1
    Worker 1: actual time=0.012..0.026 rows=2.00 loops=1
        Buffers: shared hit=1
-> Seq Scan on public.region r (cost=0.00..20.62 rows=4 width=4) (actual time=0.005..0.239 rows=2.00 loops=3)
    Output: r.r_regionkey
    Filter: (r.r_name = 'EUROPE'::text)
    Rows Removed by Filter: 8
    Buffers: shared hit=3
    Worker 0: actual time=0.003..0.690 rows=2.00 loops=1
        Buffers: shared hit=1
    Worker 1: actual time=0.003..0.017 rows=2.00 loops=1
        Buffers: shared hit=1
-> Hash (cost=6667.47..6667.47 rows=876 width=19) (actual time=0.8765382..0.87675792 rows=500.00 loops=1)
    Output: part.p_partkey, part.p_mfgr
    Buckets: 1024 Batches: 1 Memory Usage: 33kB
    Buffers: shared hit=669170 read=16043902
-> Gather (cost=1000.00..6667.47 rows=876 width=19) (actual time=1.766..4.405 rows=820.00 loops=1)
    Output: part.p_partkey, part.p_mfgr
    Workers Planned: 2
    Workers Launched: 2
    Buffers: shared hit=334 read=3882
-> Parallel Seq Scan on public.part (cost=0.00..5579.87 rows=365 width=19) (actual time=2.250..48.062 rows=273.33 loops=3)
    Output: part.p_partkey, part.p_mfgr
    Filter: ((part.p_type ~ '%BRASS'::text) AND (part.p_size = 15))
    Rows Removed by Filter: 73060
    Buffers: shared hit=334 read=3882
    Worker 0: actual time=3.988..71.076 rows=407.00 loops=1
        Buffers: shared hit=201 read=1837
    Worker 1: actual time=1.210..70.460 rows=411.00 loops=1
        Buffers: shared hit=133 read=2031
SubPlan 1
-> Aggregate (cost=34072.72..34072.73 rows=1 width=8) (actual time=179.237..179.237 rows=1.00 loops=3260)
    Output: min(ps_1.ps_supplycost)
    Buffers: shared hit=2595600 read=63164624
-> Nested Loop (cost=84.74..34072.72 rows=1 width=8) (actual time=176.107..179.233 rows=7.47 loops=3260)
    Output: ps_1.ps_supplycost
    Join Filter: (nation_1.n_regionkey = region_1.r_regionkey)
    Buffers: shared hit=2595600 read=63164624
-> Nested Loop (cost=84.74..34051.84 rows=4 width=12) (actual time=176.106..179.230 rows=3.73 loops=3260)
    Output: ps_1.ps_supplycost, nation_1.n_regionkey
    Join Filter: (n_3.n_nationkey = nation_1.n_nationkey)
    Rows Removed by Join Filter: 90
    Buffers: shared hit=2595600 read=63164624
-> Nested Loop Semi Join (cost=84.74..34023.62 rows=1 width=16) (actual time=176.097..179.210 rows=1.87 loops=3260)
    Output: ps_1.ps_supplycost, s_3.s_nationkey, n_3.n_nationkey
    Join Filter: (ps_1.ps_suppkey = s_2.s_suppkey)
    Rows Removed by Join Filter: 1036
    Buffers: shared hit=2589572 read=63164624
-> Nested Loop (cost=42.37..33497.06 rows=1 width=24) (actual time=175.583..178.114 rows=1.87 loops=3260)
    Output: ps_1.ps_supplycost, ps_1.ps_suppkey, s_3.s_nationkey, s_3.s_suppkey, n_3.n_nationkey
    Join Filter: (ps_1.ps_suppkey = s_3.s_suppkey)
    Rows Removed by Join Filter: 10106
    Buffers: shared hit=2172298 read=63164624
-> Hash Join (cost=42.37..486.20 rows=862 width=12) (actual time=0.003..2.193 rows=1994.00 loops=3260)
    Output: s_3.s_nationkey, s_3.s_suppkey, n_3.n_nationkey
    Inner Unique: true
    Hash Cond: (s_3.s_nationkey = n_3.n_nationkey)
    Buffers: shared hit=841082
-> Seq Scan on public.supplier s_3 (cost=0.00..405.94 rows=10780 width=8) (actual time=0.001..1.474 rows=10026.00 loops=3260)
    Output: s_3.s_suppkey, s_3.s_name, s_3.s_address, s_3.s_nationkey, s_3.s_phone, s_3.s_acctbal, s_3.s_comment
    Filter: (s_3.s_acctbal > '0'::double precision)
    Rows Removed by Filter: 974
    Buffers: shared hit=841080
-> Hash (cost=42.17..42.17 rows=16 width=4) (actual time=0.058..0.432 rows=5.00 loops=1)
    Output: n_3.n_nationkey
    Buckets: 1024 Batches: 1 Memory Usage: 9kB
    Buffers: shared hit=2
-> HashAggregate (cost=42.01..42.17 rows=16 width=4) (actual time=0.052..0.425 rows=5.00 loops=1)

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Output: n_3.n_nationkey
Group Key: n_3.n_nationkey
Batches: 1 Memory Usage: 32kB
Buffers: shared hit=2
-> Hash Join (cost=20.68..41.97 rows=16 width=4) (actual time=0.032..0.416 rows=20.00 loops=1)
Output: n_3.n_nationkey
Hash Cond: (n_3.n_regionkey = r_3.r_regionkey)
Buffers: shared hit=2
-> Seq Scan on public.nation n_3 (cost=0.00..18.10 rows=810 width=8) (actual time=0.005..0.009 rows=50.00 loops=1)
Output: n_3.n_nationkey, n_3.n_name, n_3.n_regionkey, n_3.n_comment
Buffers: shared hit=1
-> Hash (cost=20.62..20.62 rows=4 width=4) (actual time=0.015..0.015 rows=2.00 loops=1)
Output: r_3.r_regionkey
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=1
-> Seq Scan on public.region r_3 (cost=0.00..20.62 rows=4 width=4) (actual time=0.006..0.007 rows=2.00 loops=1)
Output: r_3.r_regionkey
Filter: (r_3.r_name = 'EUROPE'::text)
Rows Removed by Filter: 8
Buffers: shared hit=1
-> Materialize (cost=0.00..32959.15 rows=4 width=12) (actual time=0.034..0.088 rows=5.07 loops=6500440)
Output: ps_1.ps_supplycost, ps_1.ps_suppkey
Storage: Memory Maximum Storage: 17kB
Buffers: shared hit=1331216 read=63164624
-> Seq Scan on public.partsupp ps_1 (cost=0.00..32959.13 rows=4 width=12) (actual time=67.074..174.641 rows=5.07 loops=3260)
Output: ps_1.ps_supplycost, ps_1.ps_suppkey
Filter: ((ps_1.ps_availqty > 100) AND (part.p_partkey = ps_1.ps_partkey))
Rows Removed by Filter: 879995
Buffers: shared hit=1331216 read=63164624
-> Hash Join (cost=42.37..515.78 rows=862 width=4) (actual time=0.003..0.558 rows=556.07 loops=6087)
Output: s_2.s_suppkey
Inner Unique: true
Hash Cond: (s_2.s_nationkey = n_2.n_nationkey)
Buffers: shared hit=417274
-> Seq Scan on public.supplier s_2 (cost=0.00..435.52 rows=10780 width=8) (actual time=0.001..0.369 rows=2811.81 loops=6087)
Output: s_2.s_suppkey, s_2.s_name, s_2.s_address, s_2.s_nationkey, s_2.s_phone, s_2.s_acctbal, s_2.s_comment
Filter: ((s_2.s_acctbal > '0'::double precision) AND (s_2.s_acctbal > '0'::double precision))
Rows Removed by Filter: 276
Buffers: shared hit=417272
-> Hash (cost=42.17..42.17 rows=16 width=4) (actual time=0.049..0.050 rows=5.00 loops=1)
Output: n_2.n_nationkey
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=2
-> HashAggregate (cost=42.01..42.17 rows=16 width=4) (actual time=0.043..0.045 rows=5.00 loops=1)
Output: n_2.n_nationkey
Group Key: n_2.n_nationkey
Batches: 1 Memory Usage: 32kB
Buffers: shared hit=2
-> Hash Join (cost=20.68..41.97 rows=16 width=4) (actual time=0.028..0.036 rows=20.00 loops=1)
Output: n_2.n_nationkey
Hash Cond: (n_2.n_regionkey = r_2.r_regionkey)
Buffers: shared hit=2
-> Seq Scan on public.nation n_2 (cost=0.00..18.10 rows=810 width=8) (actual time=0.004..0.006 rows=50.00 loops=1)
Output: n_2.n_nationkey, n_2.n_name, n_2.n_regionkey, n_2.n_comment
Buffers: shared hit=1
-> Hash (cost=20.62..20.62 rows=4 width=4) (actual time=0.017..0.017 rows=2.00 loops=1)
Output: r_2.r_regionkey
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=1
-> Seq Scan on public.region r_2 (cost=0.00..20.62 rows=4 width=4) (actual time=0.005..0.006 rows=2.00 loops=1)
Output: r_2.r_regionkey
Filter: (r_2.r_name = 'EUROPE'::text)
Rows Removed by Filter: 8
Buffers: shared hit=1
-> Seq Scan on public.nation nation_1 (cost=0.00..18.10 rows=810 width=8) (actual time=0.002..0.004 rows=50.00 loops=6087)
Output: nation_1.n_nationkey, nation_1.n_name, nation_1.n_regionkey, nation_1.n_comment
Buffers: shared hit=6087
-> Materialize (cost=0.00..20.64 rows=4 width=4) (actual time=0.000..0.000 rows=2.00 loops=12174)
Output: region_1.r_regionkey
Storage: Memory Maximum Storage: 17kB
Buffers: shared hit=1
-> Seq Scan on public.region region_1 (cost=0.00..20.62 rows=4 width=4) (actual time=0.002..0.003 rows=2.00 loops=1)
Output: region_1.r_regionkey
Filter: (region_1.r_name = 'EUROPE'::text)
Rows Removed by Filter: 8
Buffers: shared hit=1

```

```

Planning:
  Buffers: shared hit=252
Planning Time: 41.099 ms
Execution Time: 584614.410 ms
(367 rows)

```

tpch=>